

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Zoology

Paper: ZOB-101 — Animal Diversity & Fundamentals of Cell Biology
(Old Course 2012-13 Syllabus)

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A
(Animal Diversity) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2–3 sentences: [3 × 2 = 6]
- (i) Protozoans belong to subphylum Cnidospora and Sporozoa produce same kind of spores.
 - (ii) The secondary host of *Fasciola hepatica* is sheep.
 - (iii) The lung fish belong to subclass Sarcopterygii.
- (b) Differentiate between the following: [2 × 2 = 4]
- (i) Homology and analogy
 - (ii) Flightless and flying birds.
- (c) Define metamerism. [1]

Question 2. Answer the following parts: [7 + 5 = 12]

- (a) Classify phylum Echinodermata up to classes giving characteristic features and examples of each class.
- (b) Describe different types of scales in fishes.

Question 3. Answer the following parts: [7 + 5 = 12]

- (a) Give an account of salient features of phylum Porifera. Classify up to classes giving characteristic features and examples of each class.
- (b) Describe the canal system in Phylum Porifera.

Question 4. Write short notes on any three of the following: [3 × 4 = 12]

- (a) Air sacs in birds
- (b) Protochordates
- (c) Chiroptera
- (d) Sarcomastigophora.

SECTION – B
(Fundamentals of Cell Biology) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Write whether the following statements are True or False, giving reasons in 2–3 sentences:
[2 × 4 = 8]

- (i) Eukaryotic cells have cell wall made-up of peptidoglycans.
- (ii) Cellular respiration occurs on the inner layer of plasma membrane of alveolar cells only.
- (iii) Polytene chromosomes are interphase chromosomes.
- (iv) Homozygous loss of function of an oncogene leads to the transformation of cells from normal to cancerous state.

(b) Define the following: [1 × 3 = 3]

- (i) Prokaryote
- (ii) Cell cycle
- (iii) Antibody.

Question 2. Write notes on the following: [4 × 3 = 12 (or 3 × 4 = 12)]

- (a) Function of rough endoplasmic reticulum
- (b) Image formation in a student's compound microscope
- (c) Significance of meiosis.

Question 3. Answer the following parts: [6 + 6 = 12]

- (a) Draw a labeled flow chart diagram to depict biogenesis of ribosomes (no descriptions required).
- (b) Illustrate how the sorting of lysosomal proteins takes place and discuss the function of a lysosome.

Question 4. Answer the following parts: [6 + 6 = 12]

- (a) Differentiate between mitosis and meiosis showing clearly the changes in DNA content at each phase of division.
- (b) Illustrate structure and function of peroxisome.

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